

SHETH L.U.J. AND SIR M.V. COLLEGE  
SUBJECT: Data Visualization with Power BI and Tableau

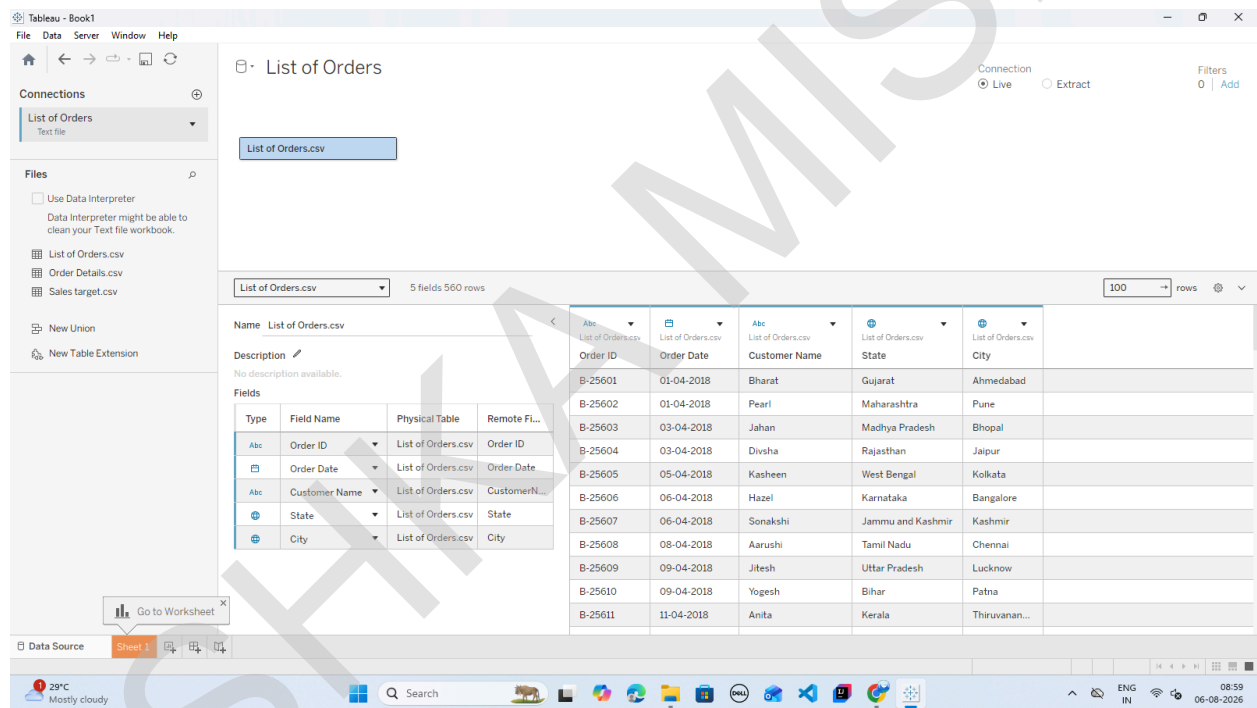
PRACTICAL NO. 6

- AIM: a) Apply worksheet-level and workbook-level formatting  
b) Customize number, date, and null value formatting  
c) Design effective color schemes and layouts  
d) Create informative tooltips  
e) Implement “Viz in Tooltip”  
f) Improve readability and user experience  
g) Apply best practices for visual storytelling

A: Apply Worksheet-Level and Workbook-Level Formatting

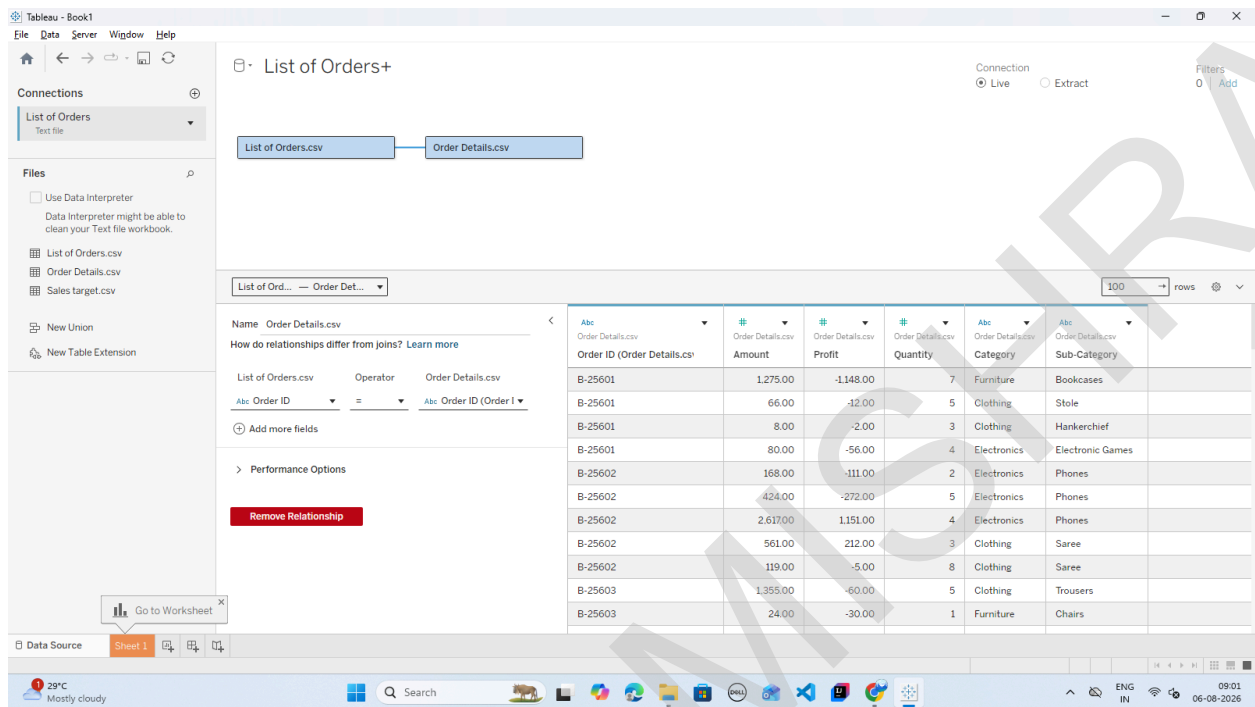
Task 1: Connect to the Dataset and Build a Baseline Chart

1. Download the Dataset:
2. Connect Data in Tableau:

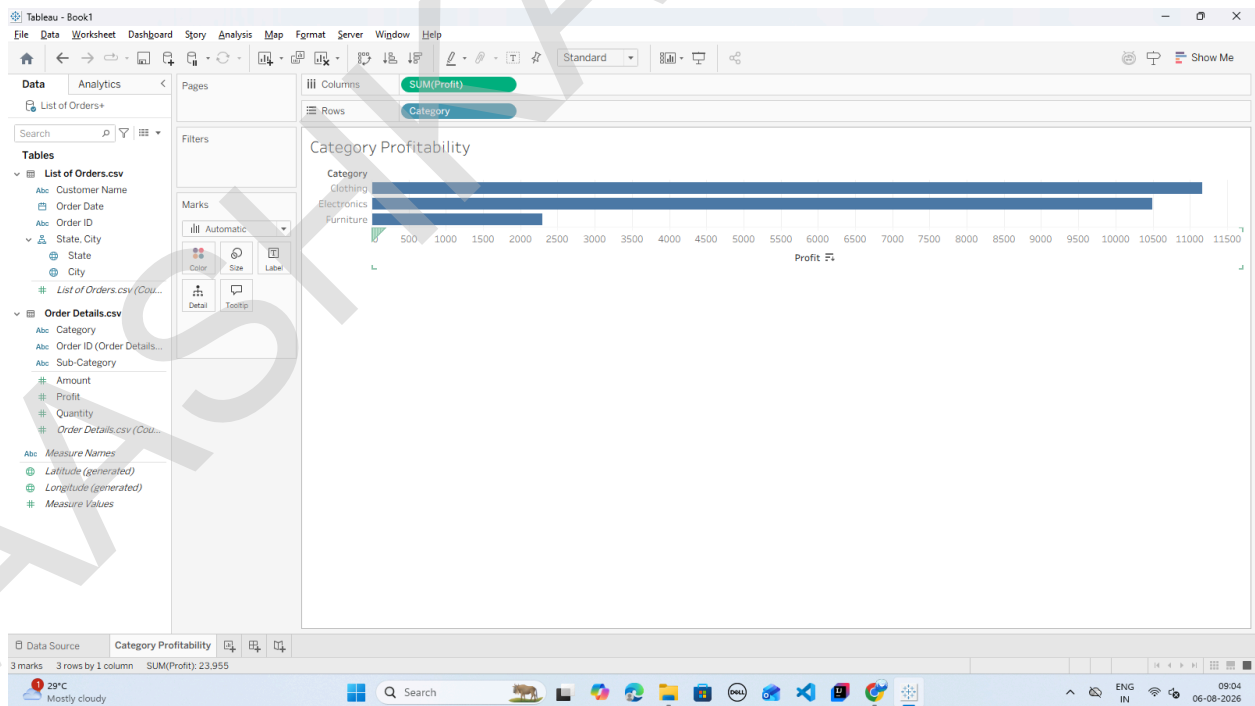


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SUBJECT: Data Visualization with Power BI and Tableau

3. Relate the Two Files:



4. Create the Base Chart:

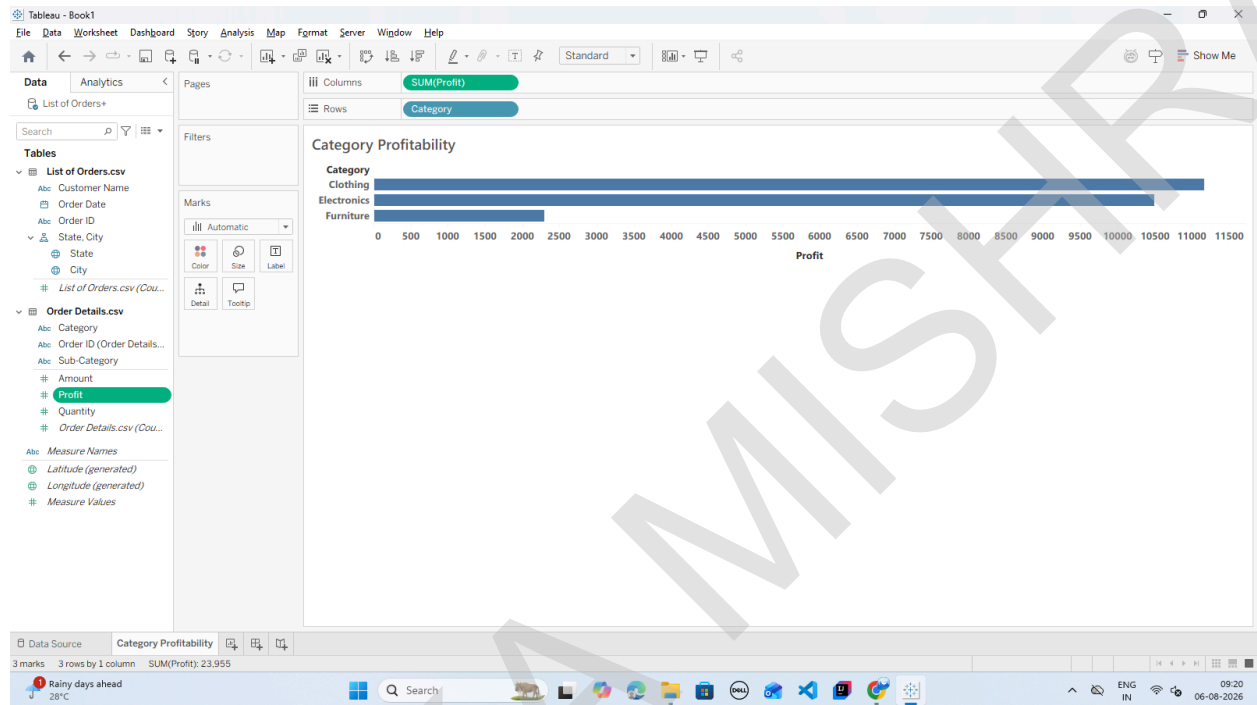


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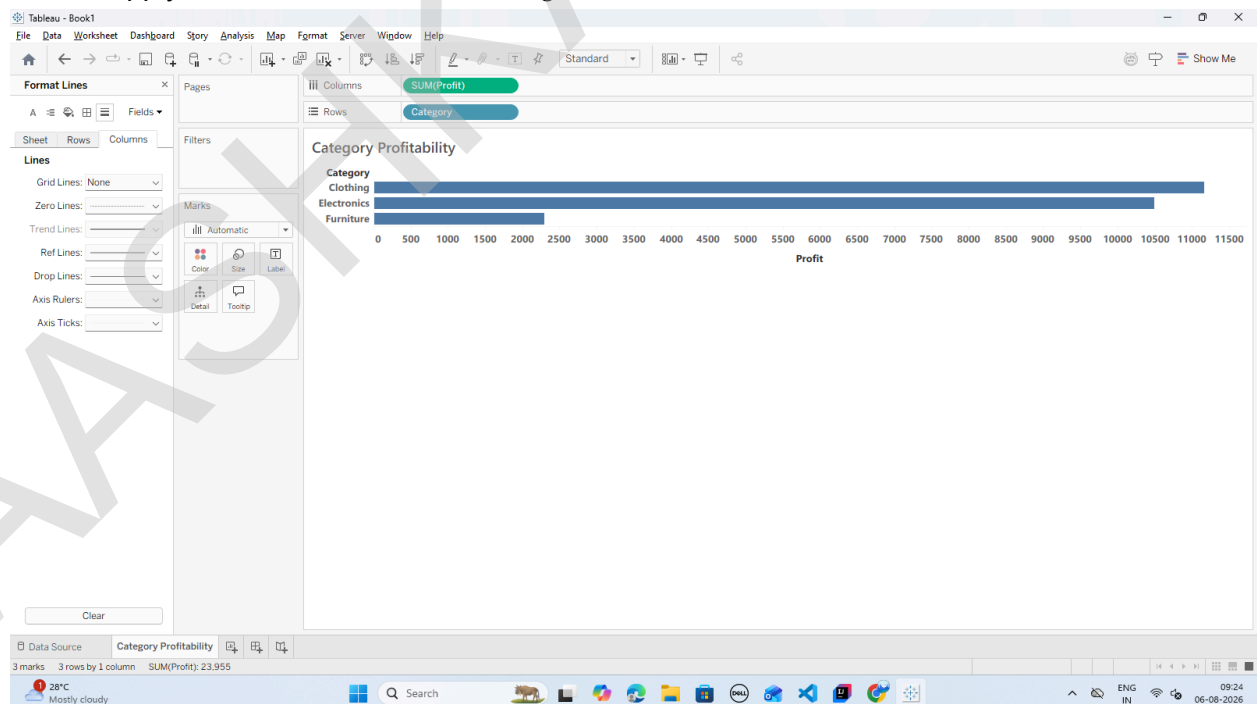
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SUBJECT: Data Visualization with Power BI and Tableau

**Task 2: Apply Workbook-Level Formatting**

1. Go to the top menu bar, click Format, then select Workbook.
2. A Format Workbook panel will open on the left side of your screen:



**Task 3: Apply Worksheet-Level Formatting**



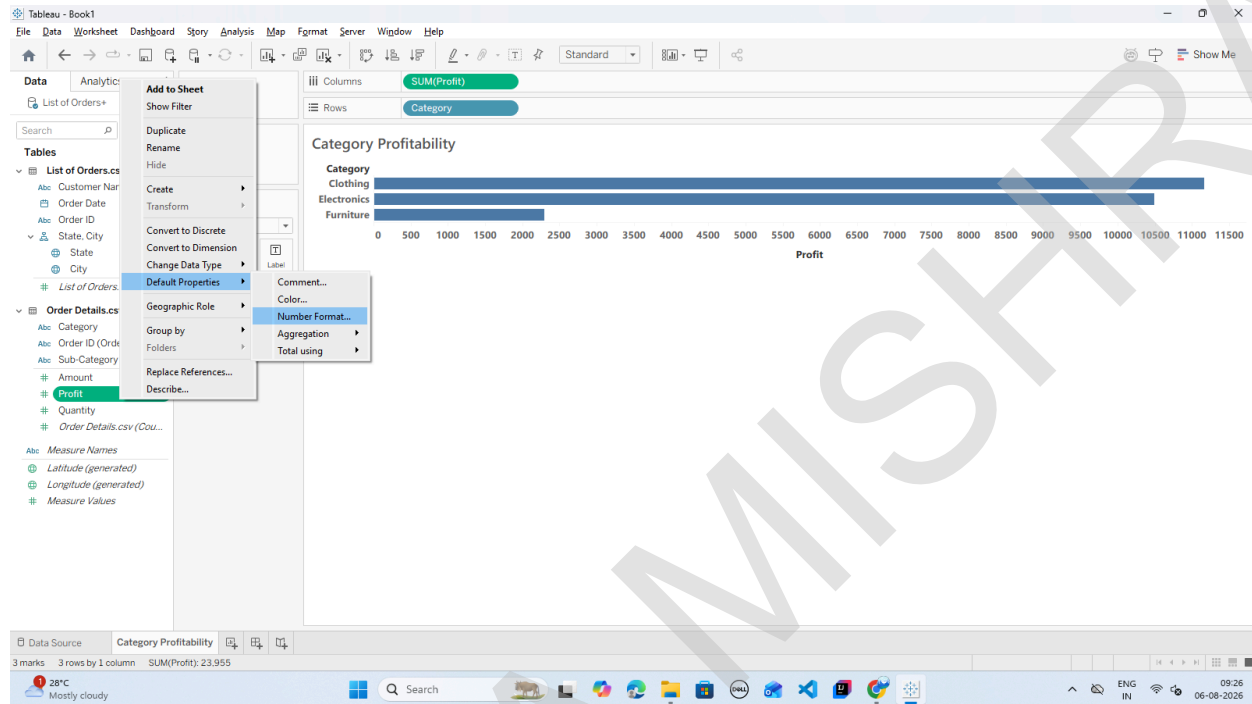
NAME: AASHKA MISHRA  
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SHETH L.U.J. AND SIR M.V. COLLEGE  
SUBJECT: Data Visualization with Power BI and Tableau

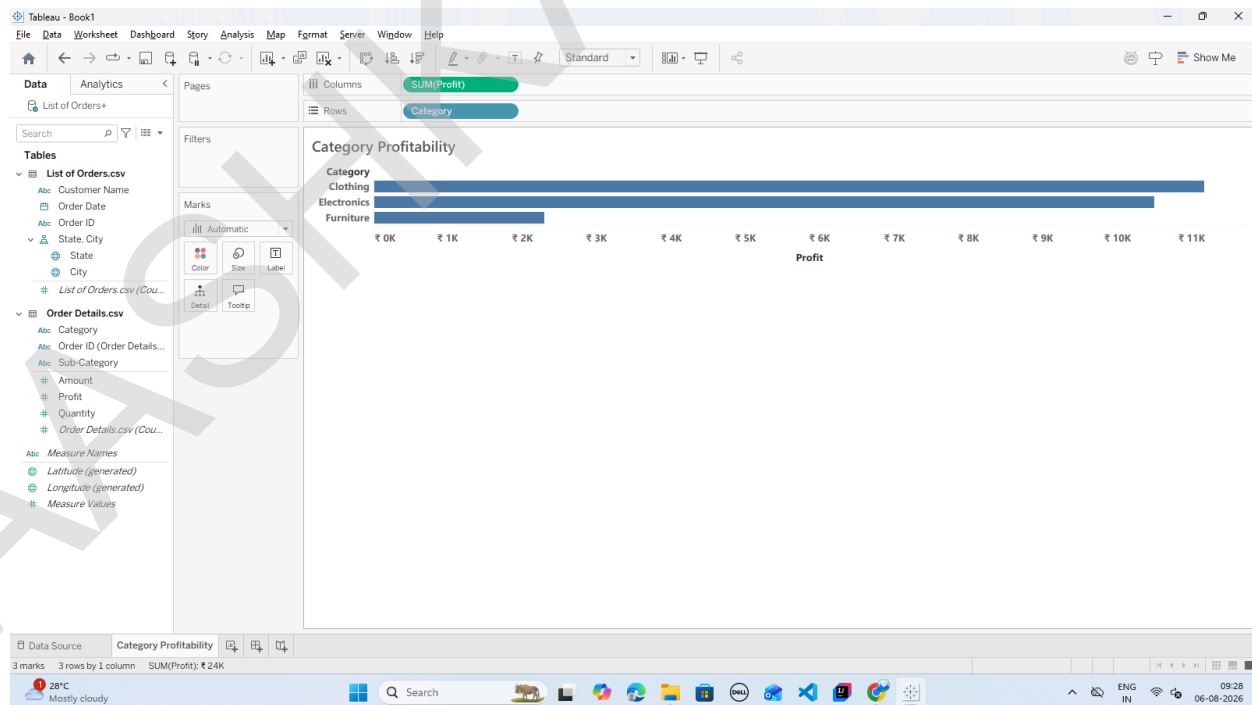
**B: Customize Number, Date, and Null Value Formatting**

**Task 1: Format Profit and Amount as Indian Currency (INR / ₹)**

1. In the Data Pane on the left side of your screen, find Profit under Order Details.
2. Right-click Profit, hover over Default Properties, and click Number Format.



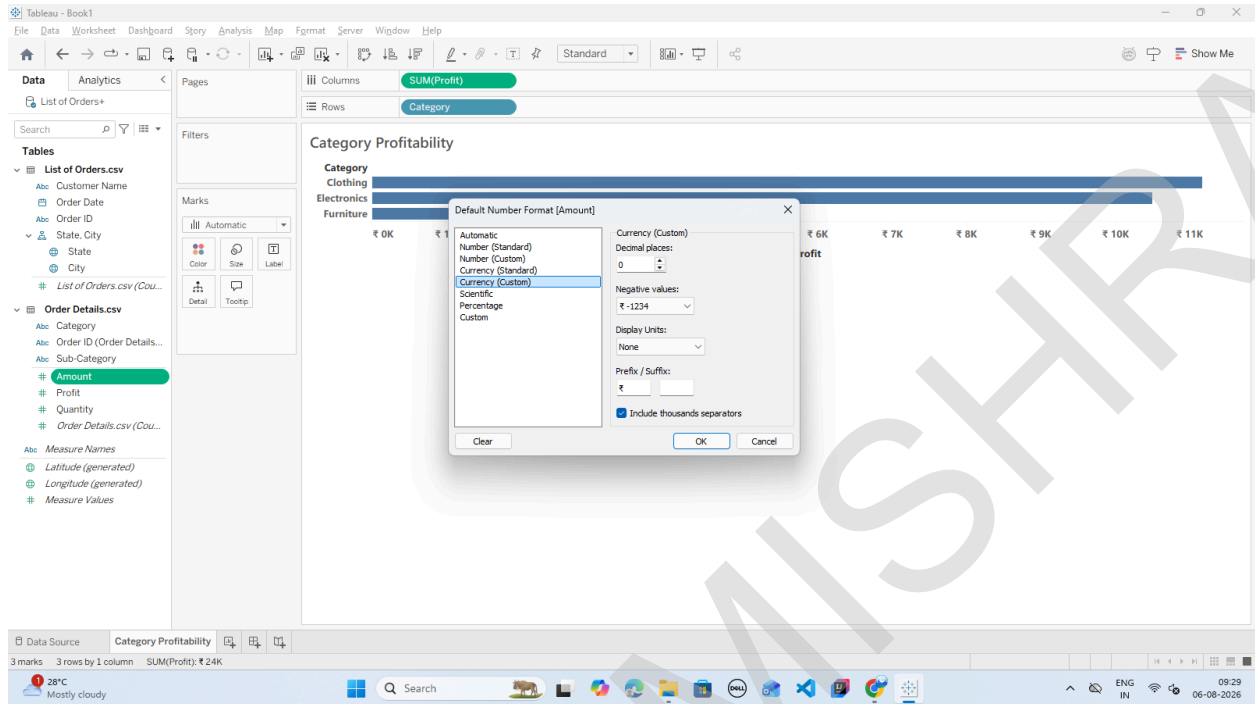
**3. In the dialog box that opens:**



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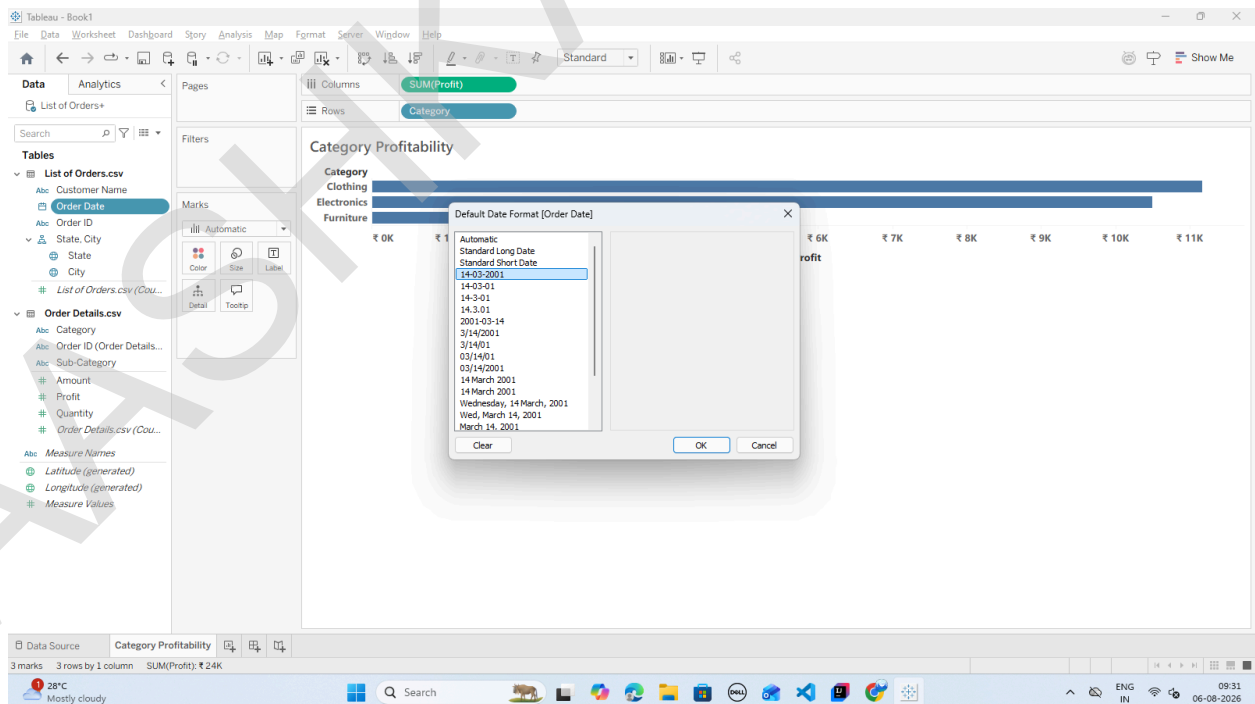
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SUBJECT: Data Visualization with Power BI and Tableau

4. Repeat these steps for the Amount field:



### Task 2: Standardize Date Formatting

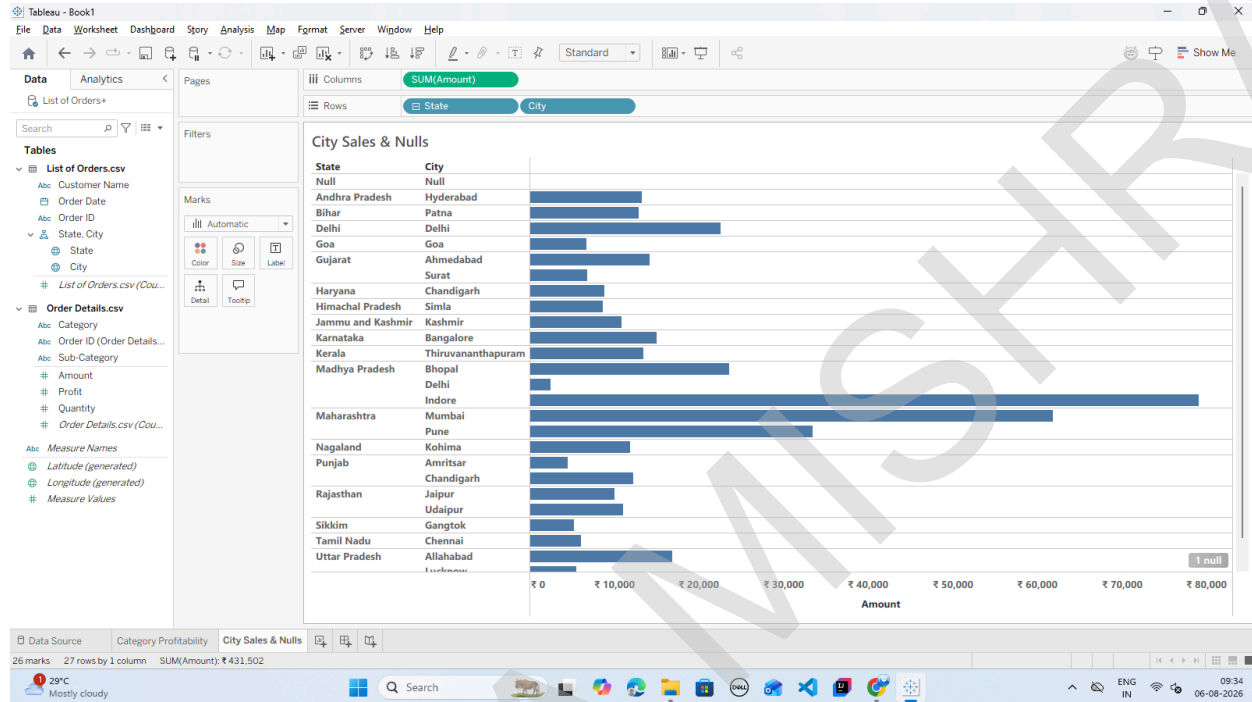
1. In the Data Pane, locate Order Date under List of Orders.
2. Right-click Order Date, hover over Default Properties, and click Date Format.
3. In the menu:



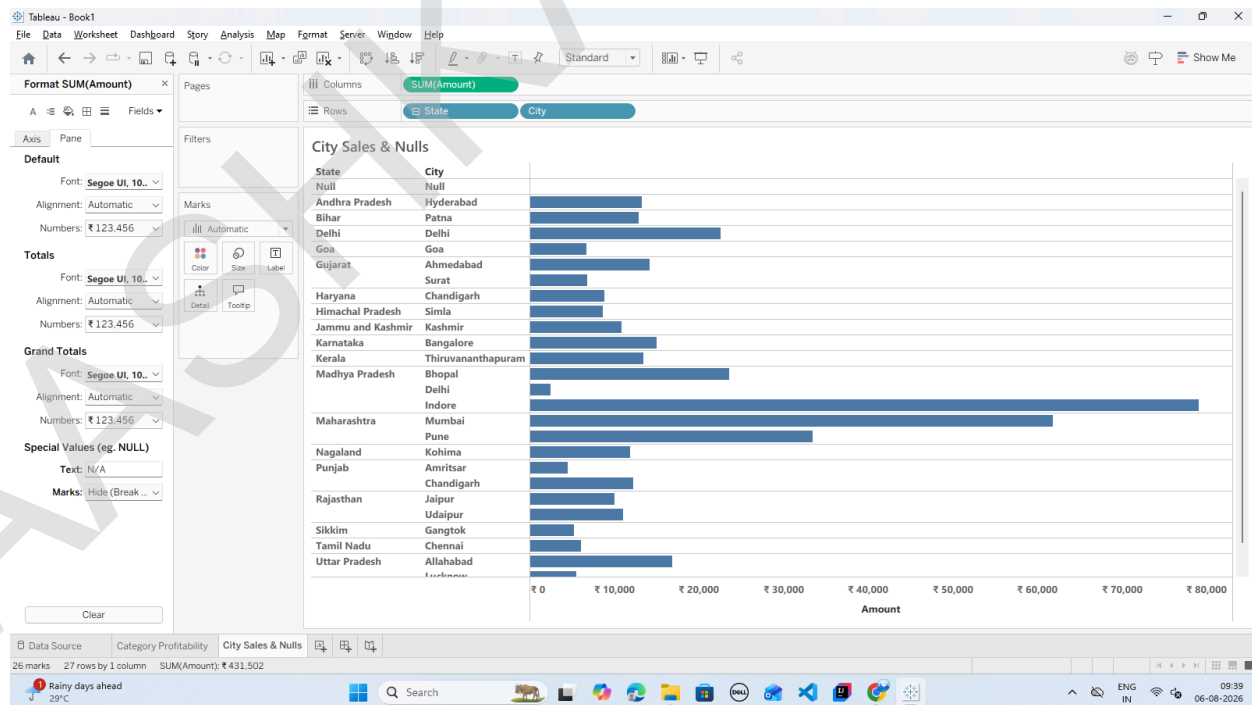
NAME: AASHKA MISHRA  
ROLL NO: T095

SHETH L.U.J. AND SIR M.V. COLLEGE  
SUBJECT: Data Visualization with Power BI and Tableau

- Task 3: Handle and Custom-Format Null Values
1. Create a new worksheet by clicking the New Worksheet tab at the bottom and name it: City Sales & Nulls.
  2. From the Data Pane, drag State to the Rows shelf.
  3. Drag City to the Rows shelf (placing it next to State).
  4. Drag Amount to the Columns shelf.



5. Notice if any rows contain blank cells or say Null.
6. Customize Null Text:



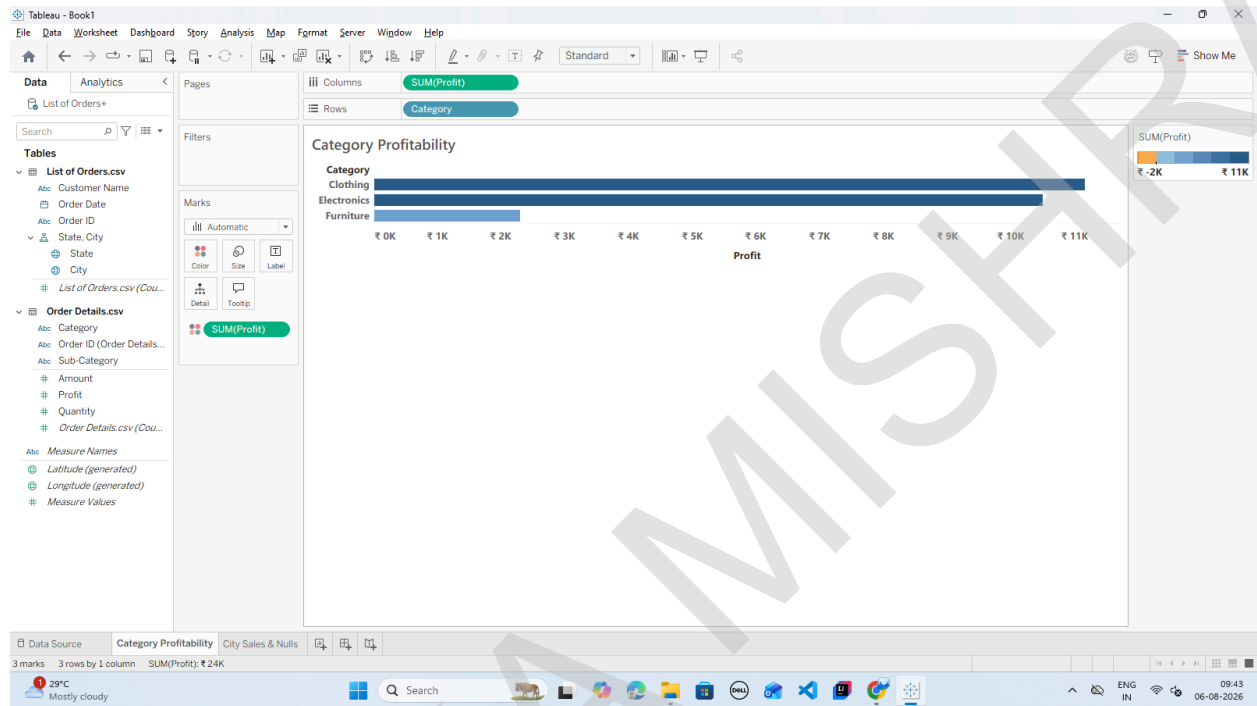
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SHETH L.U.J. AND SIR M.V. COLLEGE  
SUBJECT: Data Visualization with Power BI and Tableau

C: Design Effective Color Schemes and Layouts

Task 1: Apply a Diverging Colour Scheme for Profitability

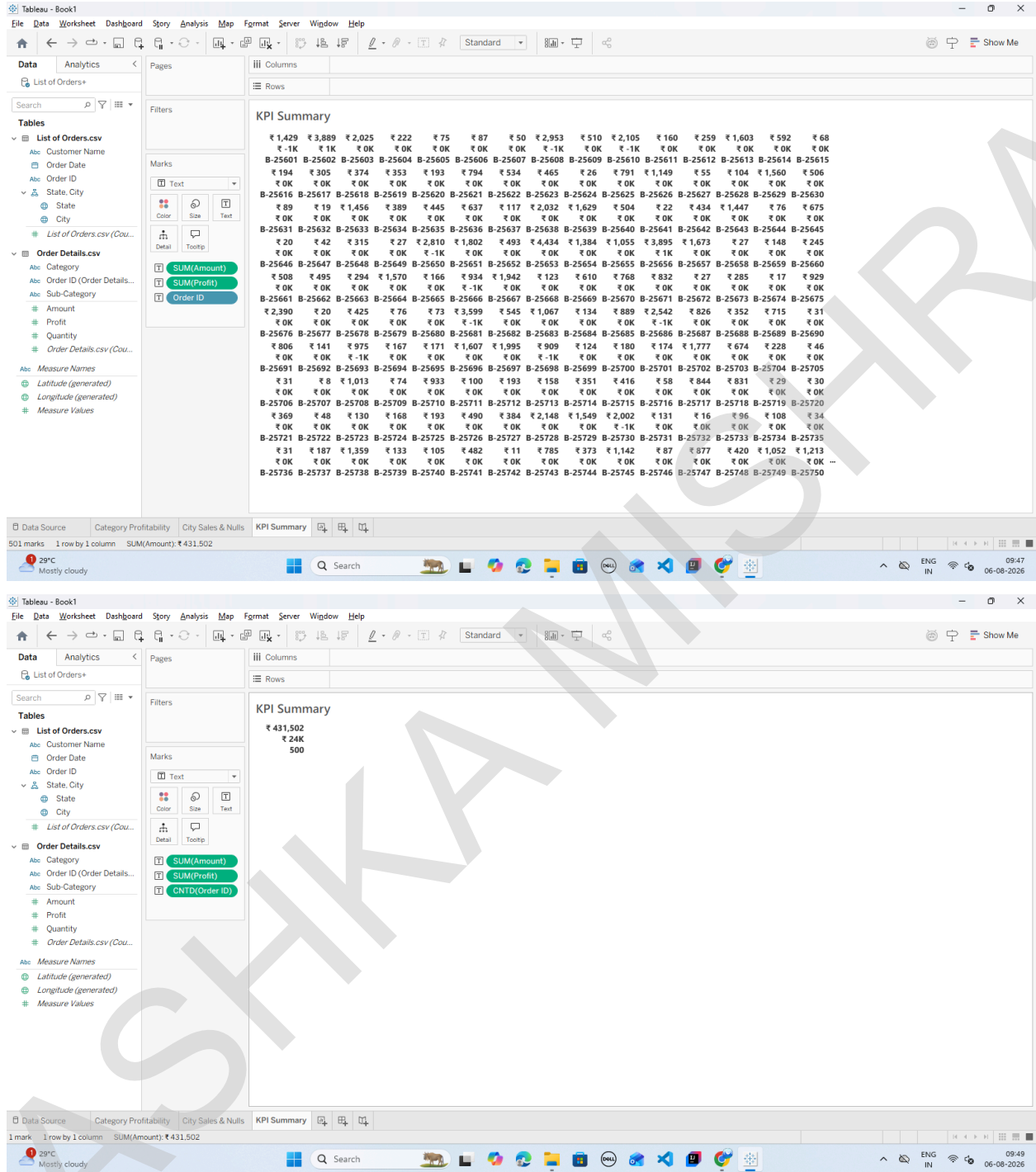
1. Click on your Category Profitability worksheet tab at the bottom.
2. From the Data Pane on the left, drag Profit directly onto the Color card in the Marks card.
3. Tableau will assign a default color gradient. To customize it:



Task 2: Build a Clean KPI Summary Sheet (Card Layout)

1. Click the New Worksheet icon at the bottom and name it: KPI Summary.
2. Change the Mark type in the Marks card drop-down from Automatic to Text.
3. Drag Amount onto the Text card (it will display as SUM(Amount)).
4. Drag Profit onto the Text card (it will display as SUM(Profit)).
5. Drag Order ID onto the Text card.

SHETH L.U.J. AND SIR M.V. COLLEGE  
SUBJECT: Data Visualization with Power BI and Tableau

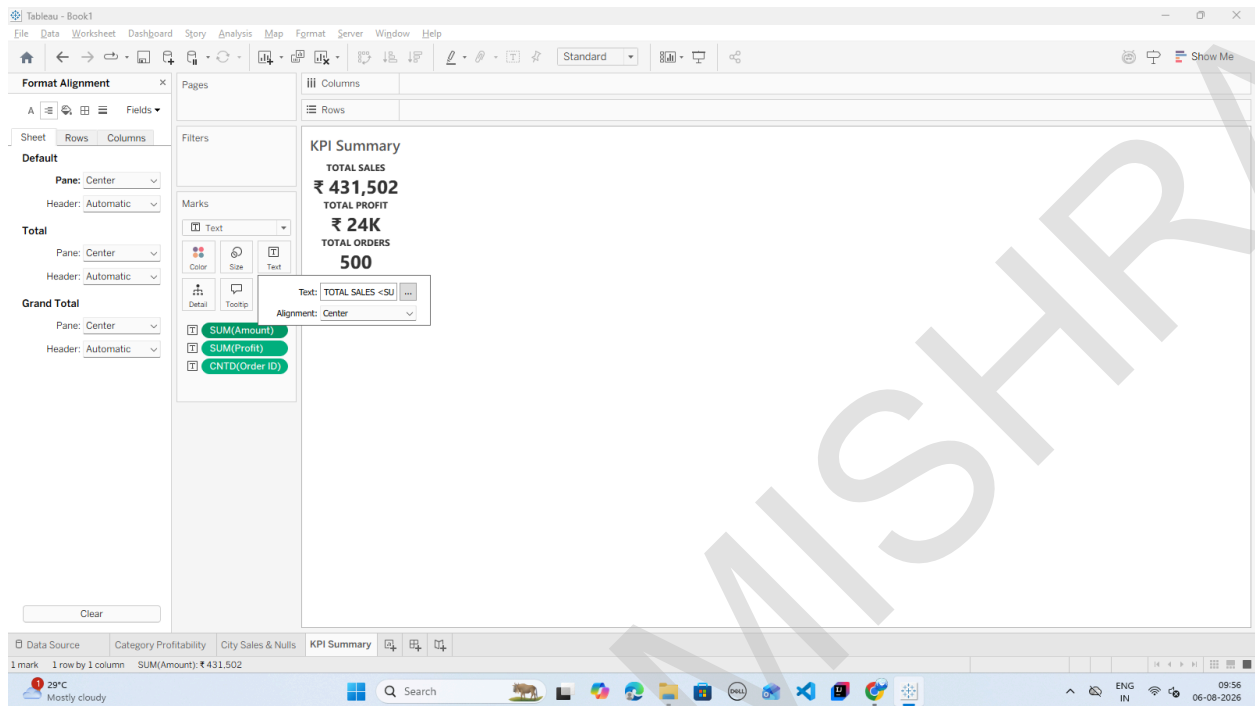


NAME: AASHKA MISHRA  
ROLL NO: T095



SHETH L.U.J. AND SIR M.V. COLLEGE  
SUBJECT: Data Visualization with Power BI and Tableau

6. Click the Text card on the Marks card, then click the ... button next to Text to edit the text layout:

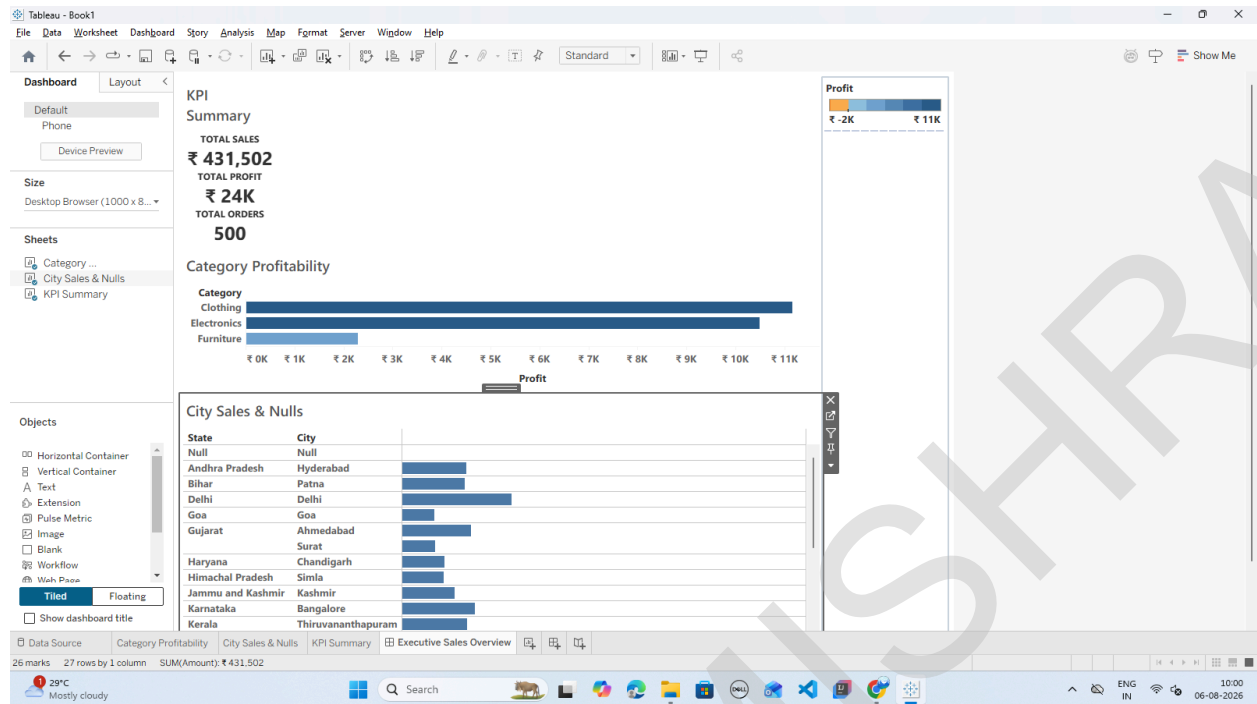


### Task 3: Create a Dashboard Container Layout

1. At the bottom of the screen, click the New Dashboard icon (the icon with a grid)
2. Rename the dashboard tab: Executive Sales Overview.
3. In the Dashboard pane on the far left:
4. Under Objects at the bottom left:
5. Drag your worksheets from the Sheets list into the container:
6. Click on the title of any sheet that isn't needed, right-click, and uncheck Title if you prefer a clean container header.

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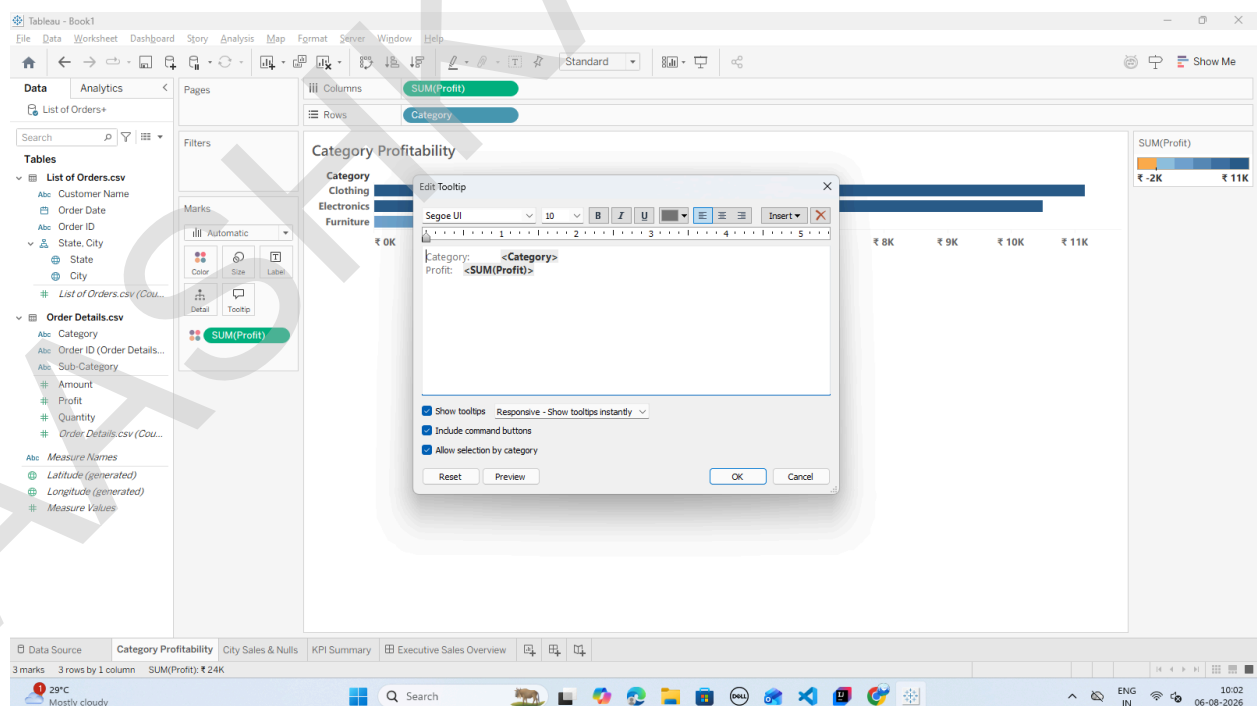
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### D: Create Informative Tooltips

#### Task 1: Open the Tooltip Editor

1. Click on your Category Profitability worksheet tab at the bottom.
2. On the Marks card (to the left of the chart), click the Tooltip card.
3. The Edit Tooltip dialog box will open showing the default Tableau code:

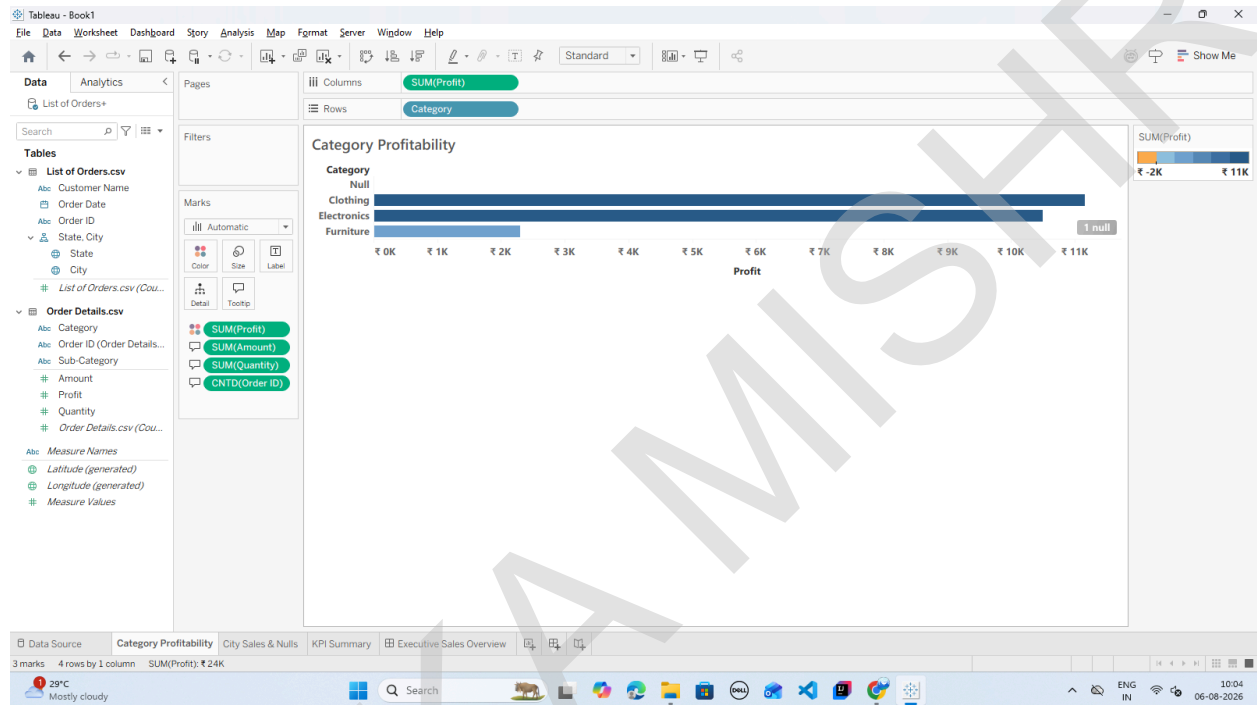


NAME: AASHKA MISHRA  
ROLL NO: T095

SHETH L.U.J. AND SIR M.V. COLLEGE  
SUBJECT: Data Visualization with Power BI and Tableau

**Task 2: Add Contextual Measures to the Tooltip**

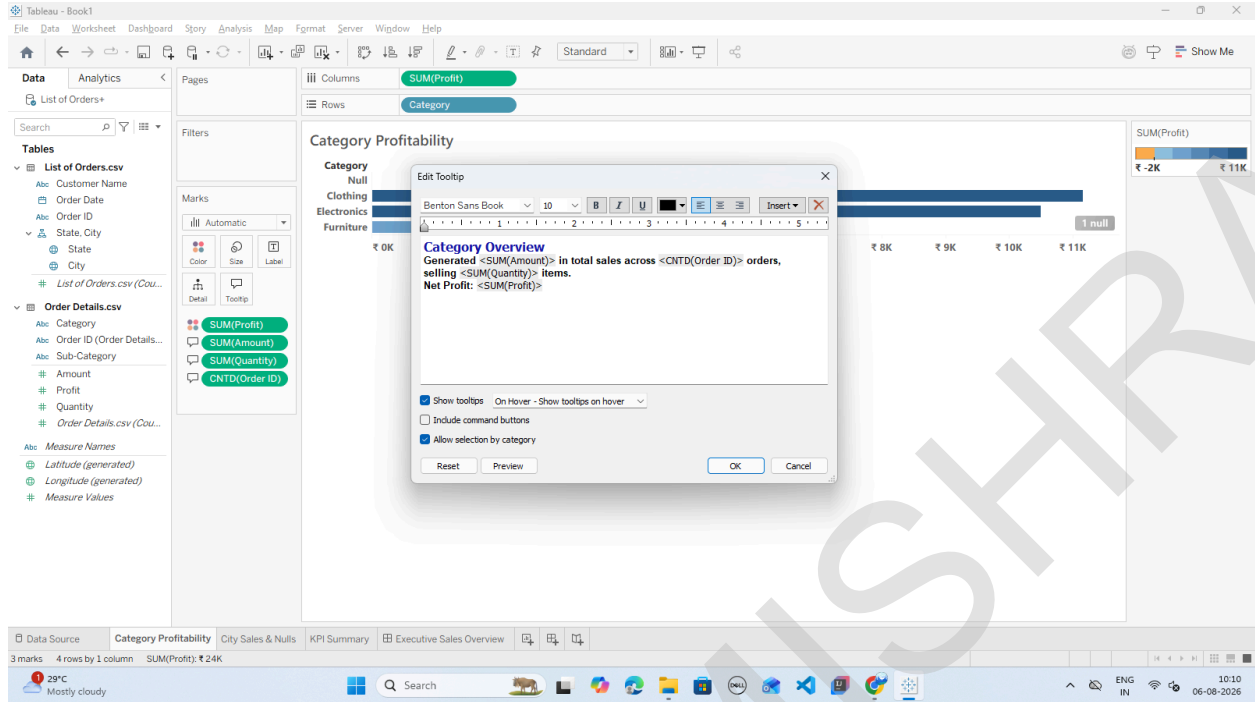
- o Click Cancel to temporarily close the Edit Tooltip window.
- o From the Data Pane, drag Amount onto the Tooltip card on the Marks card.
- o Drag Quantity onto the Tooltip card on the Marks card.
- o Drag Order ID onto the Tooltip card, then click its drop-down arrow  $\rightarrow$  Measure  $\rightarrow$  Count (Distinct) so it displays as CNTD(Order ID).



**Task 3: Format the Tooltip into a Sentence-Based Story**

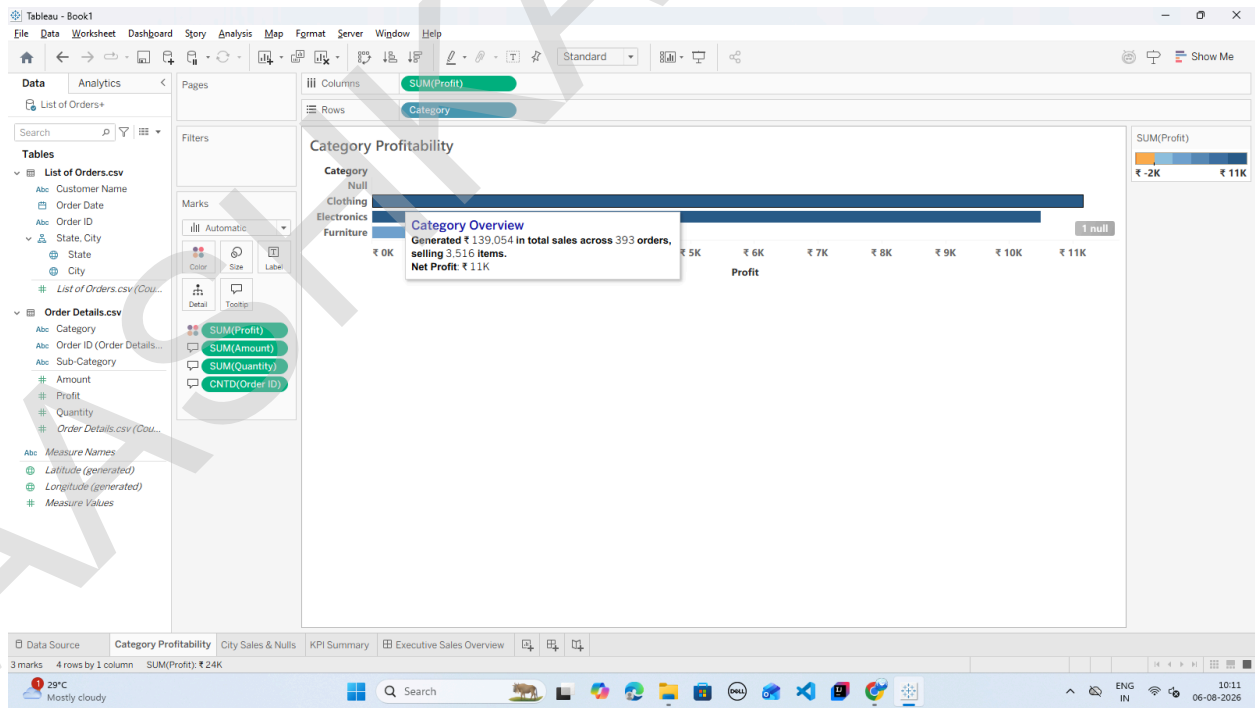
1. Click the Tooltip card on the Marks card again to open the Edit Tooltip window.
2. Delete all existing text inside the box.
3. Re-type or paste the following clean, conversational text:
4. Style the text for visual hierarchy:

SHETH L.U.J. AND SIR M.V. COLLEGE  
SUBJECT: Data Visualization with Power BI and Tableau



#### Task 4: Test Your Tooltip

1. Move your mouse cursor over any bar in the Category Profitability view.
2. Confirm that the popup reads as a clean, formatted sentence displaying exact sales, order counts, and net profit for that category!



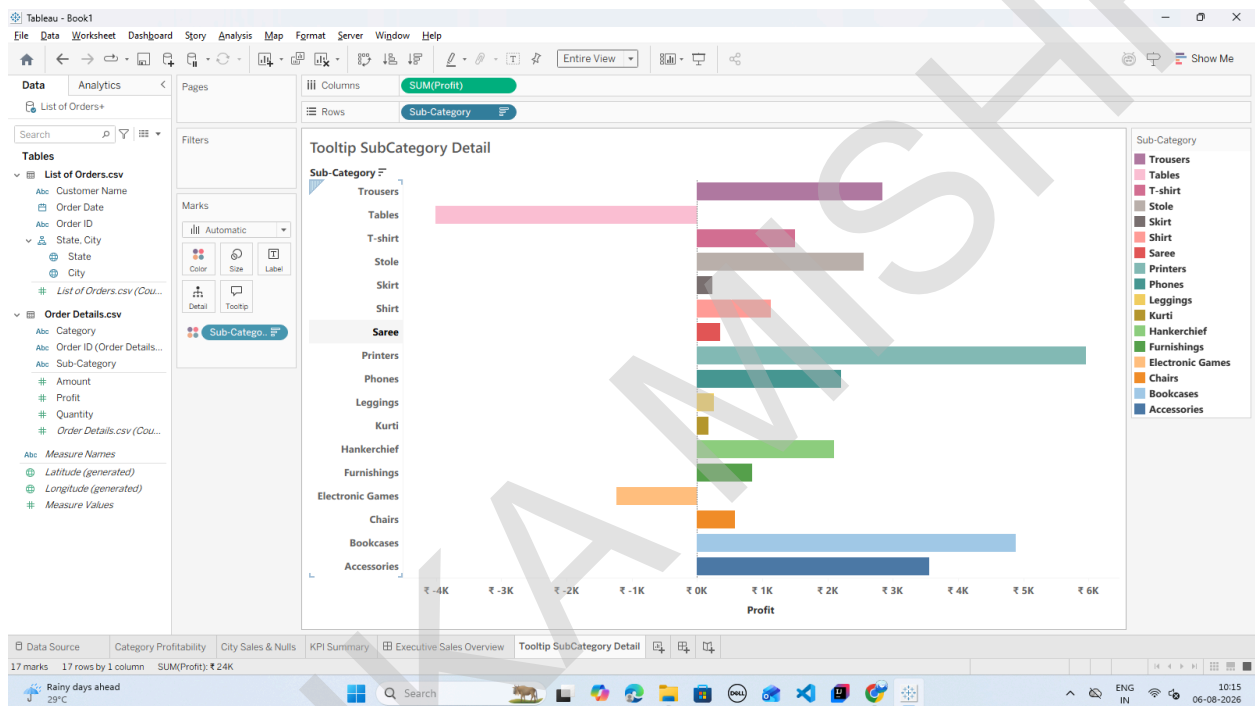
NAME: AASHKA MISHRA  
ROLL NO: T095

SHETH L.U.J. AND SIR M.V. COLLEGE  
SUBJECT: Data Visualization with Power BI and Tableau

E: Implement "Viz in Tooltip"

Task 1: Build the Secondary Detailed Chart (Target Sheet)

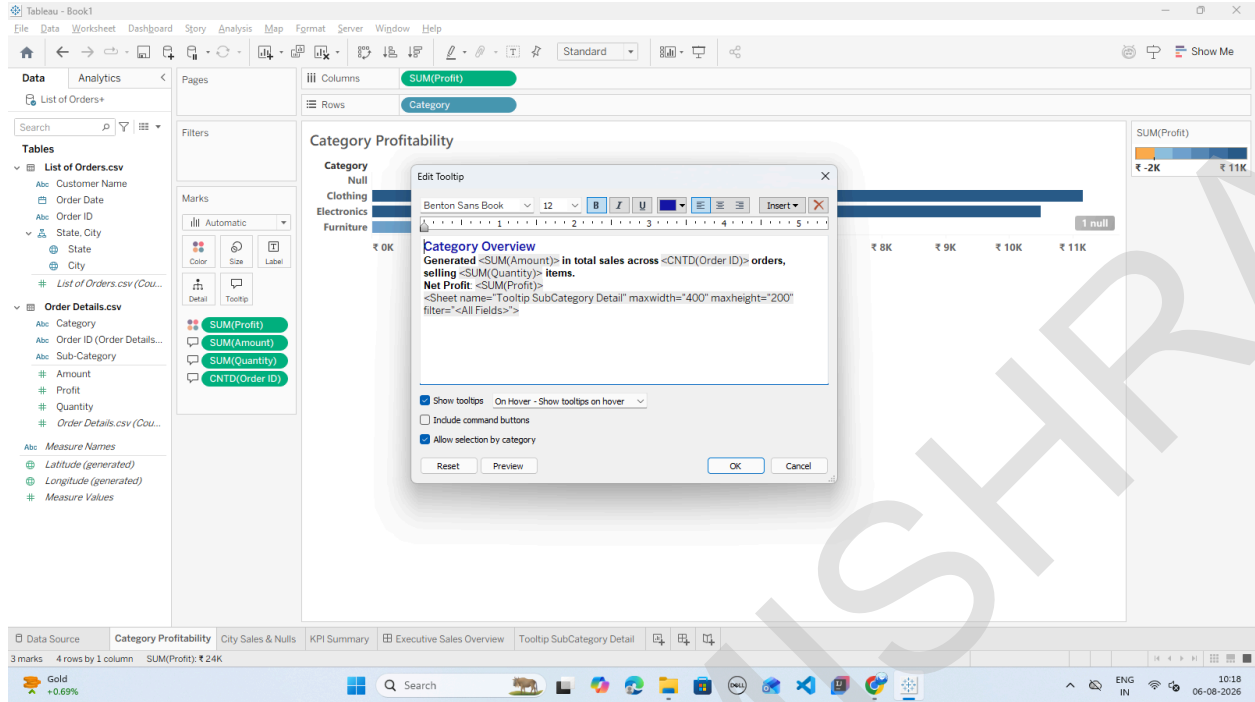
1. Click the New Worksheet icon at the bottom of the screen and name it: Tooltip SubCategory Detail.
2. From the Data Pane, drag Sub-Category to the Rows shelf.
3. Drag Profit to the Columns shelf.
4. Drag Profit to the Color card on the Marks card to give it visual context.
5. Sort the bars: Click the Sort Descending button on the top toolbar (or right-click SubCategory on Rows Sort Nested by Profit Descending).



Task 2: Insert the Detailed View into the Main Chart's Tooltip

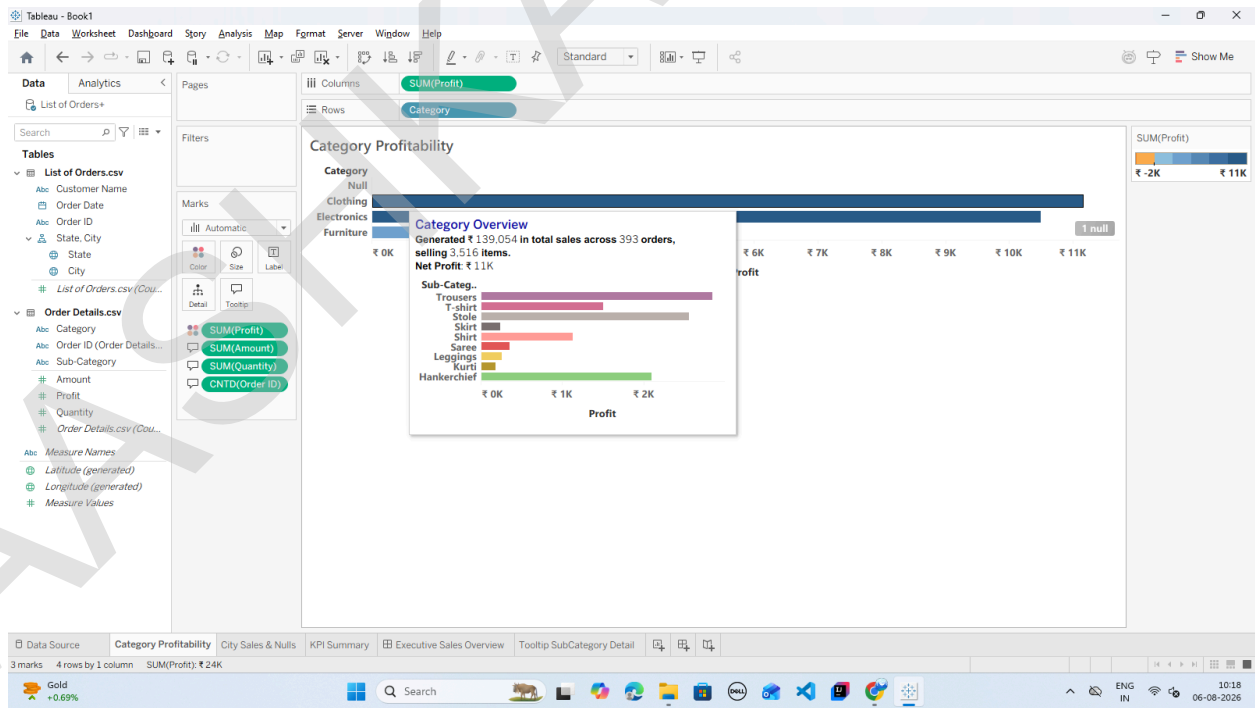
1. Click on your primary worksheet tab at the bottom: Category Profitability.
2. On the Marks card, click the Tooltip card to open the Edit Tooltip window.
3. Place your cursor at the very bottom of your existing sentence-based text and press Enter to start a new line.
4. In the top right corner of the Edit Tooltip window, click the Insert drop-down menu.
5. Hover over Sheets, then click Tooltip SubCategory Detail.
6. Tableau will insert a tag that looks like this:
7. Adjust the dimensions inside the tag if needed:

SHETH L.U.J. AND SIR M.V. COLLEGE  
SUBJECT: Data Visualization with Power BI and Tableau



Task 3: Test and Verify the Interactive Filtering

1. Hover your mouse over one of the main bars (e.g., the Clothing or Furniture category bar) in your Category Profitability worksheet.
2. Observe the tooltip popup:



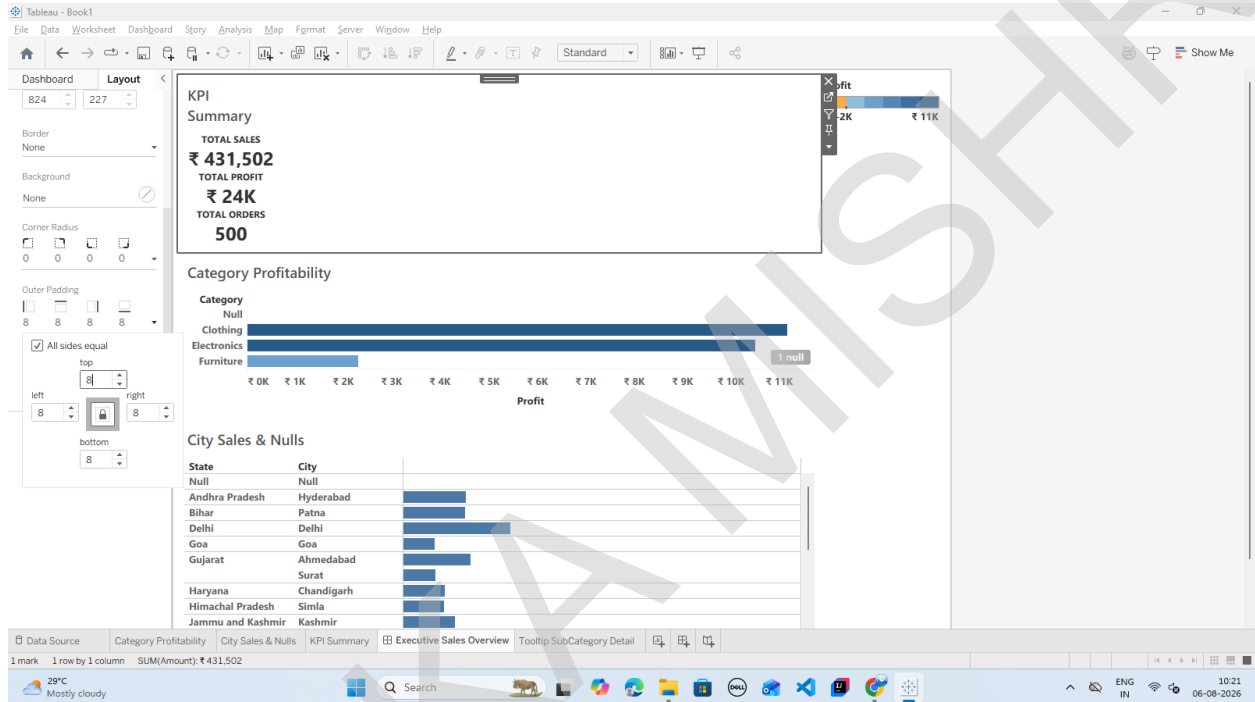
NAME: AASHKA MISHRA  
ROLL NO: T095

SHETH L.U.J. AND SIR M.V. COLLEGE  
SUBJECT: Data Visualization with Power BI and Tableau

F: Improve Readability and User Experience

Task 1: Refine Dashboard Spacing and Padding

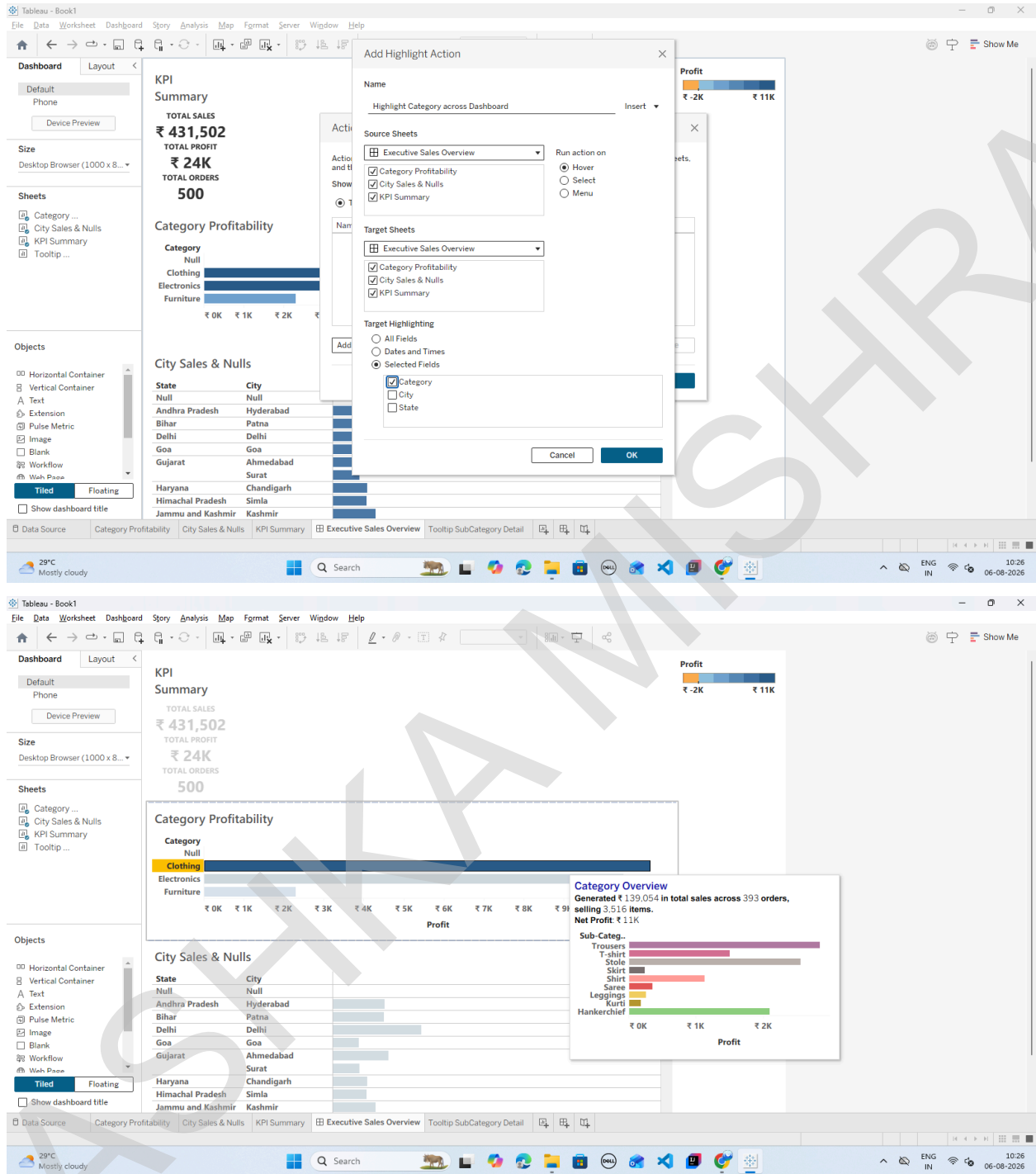
1. Switch to your Executive Sales Overview dashboard tab.
2. Click on the KPI Summary container/sheet to select it.
3. In the left-hand panel, click the Layout tab (next to the Dashboard tab).
4. Look for the Outer Padding and Inner Padding settings by scrolling:
5. Click on the Category Profitability sheet and apply an Outer Padding of 8 px.
6. Repeat for City Sales & Nulls.



Task 2: Add an Interactive Highlight Action

1. On the top menu bar, go to Dashboard Actions... (or press Ctrl + Shift + D / Cmd + Shift + D).
2. Click Add Action > at the bottom left and select Highlight.
3. In the Add Highlight Action window:

SHETH L.U.J. AND SIR M.V. COLLEGE  
SUBJECT: Data Visualization with Power BI and Tableau



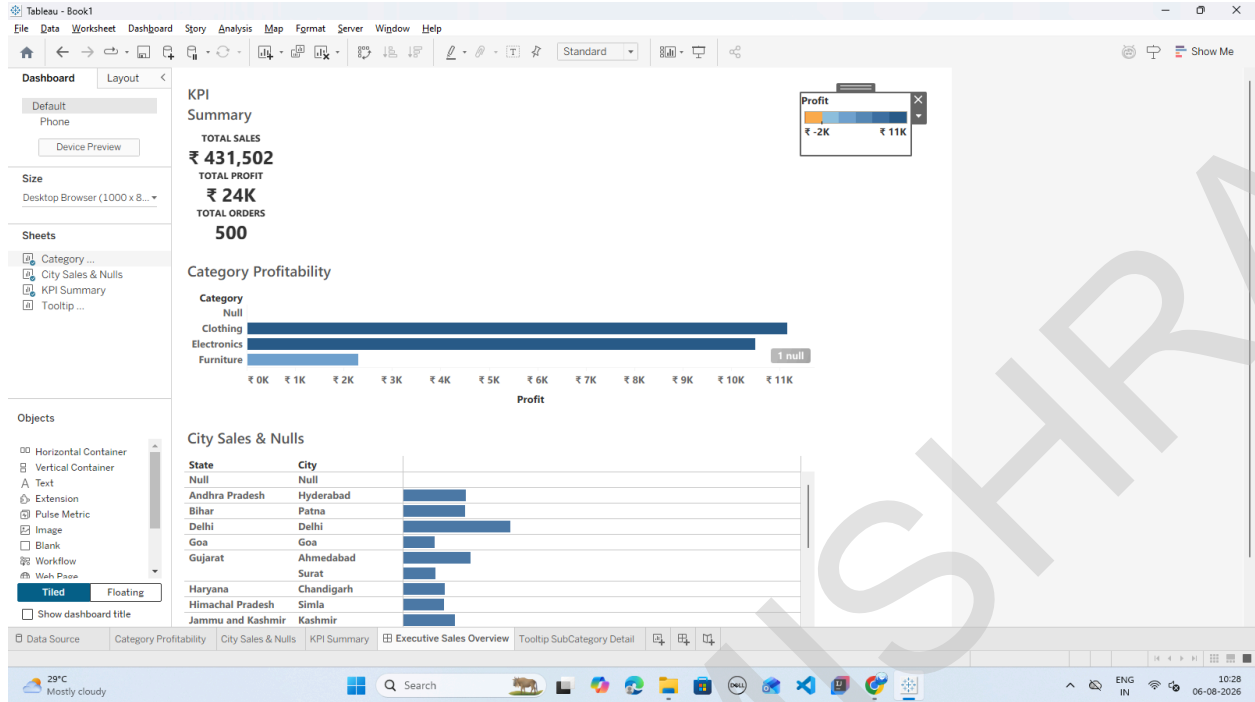
Task 3: Clean Up Color Legends and Floating Elements

1. Look at the right sidebar of your dashboard where color legends appear automatically.
2. Select any redundant or repetitive legend boxes (e.g., if Profit color legend is self-explanatory).
3. Click the X in the top right corner of the legend box to remove it, or convert critical legends into floating elements:

NAME: AASHKA MISHRA  
ROLL NO: T095

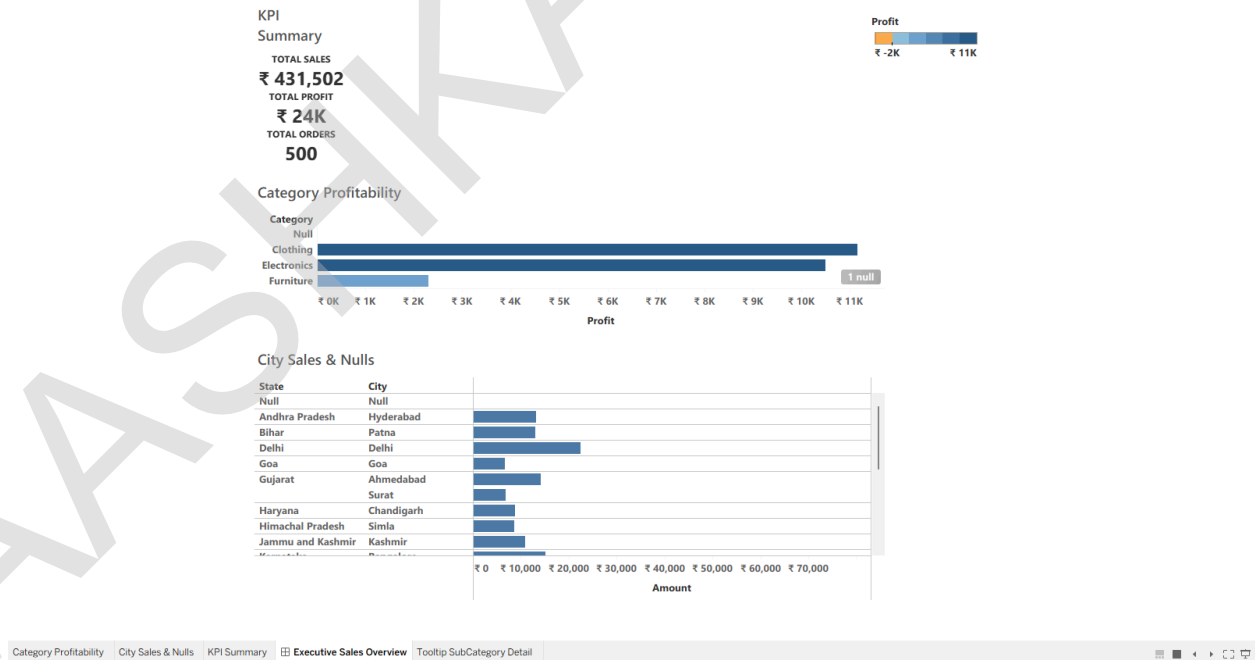


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SUBJECT: Data Visualization with Power BI and Tableau



Task 4: Final Presentation Checks

1. At the bottom right of the screen (or in the top menu under Window), click Presentation Mode (or press F5).
2. Verify:



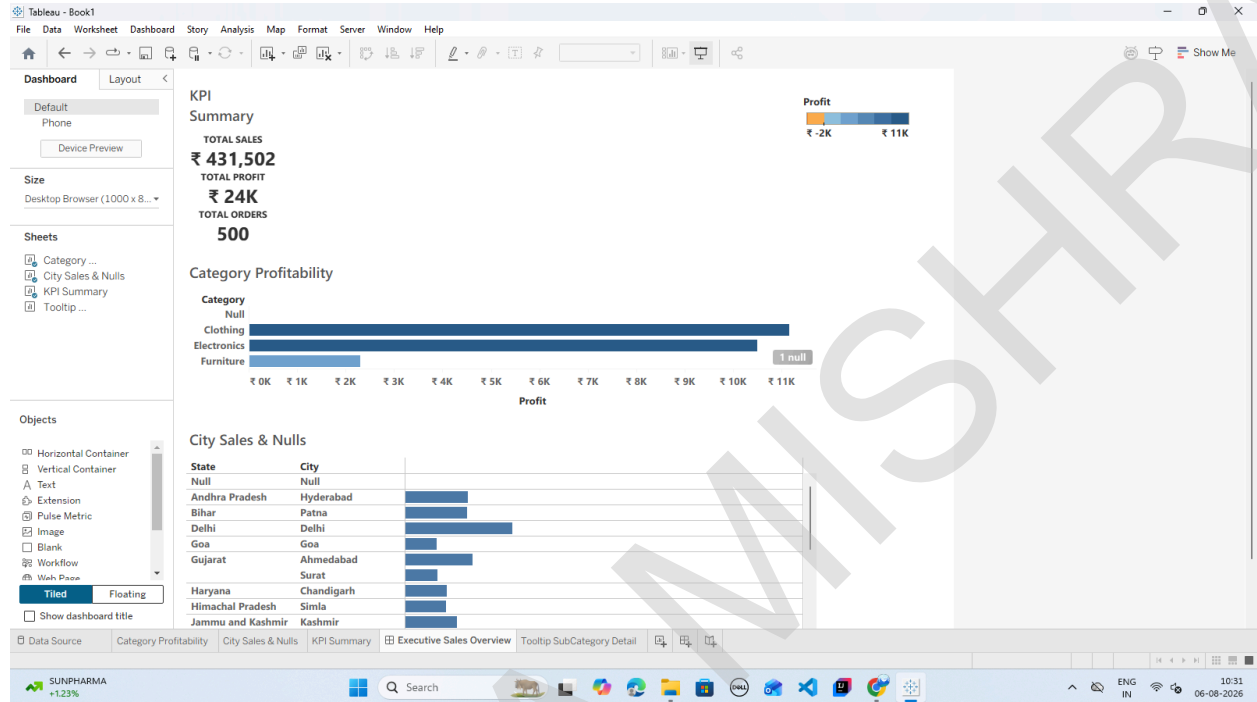
NAME: AASHKA MISHRA  
ROLL NO: T095

SHETH L.U.J. AND SIR M.V. COLLEGE  
SUBJECT: Data Visualization with Power BI and Tableau

G: Apply Best Practices for Visual Storytelling

Task 1: Re-structure Layout using the F-Pattern (Visual Hierarchy)

1. Switch to your Executive Sales Overview dashboard tab.
2. Verify or adjust the sheet placement using a top-to-bottom reading path:

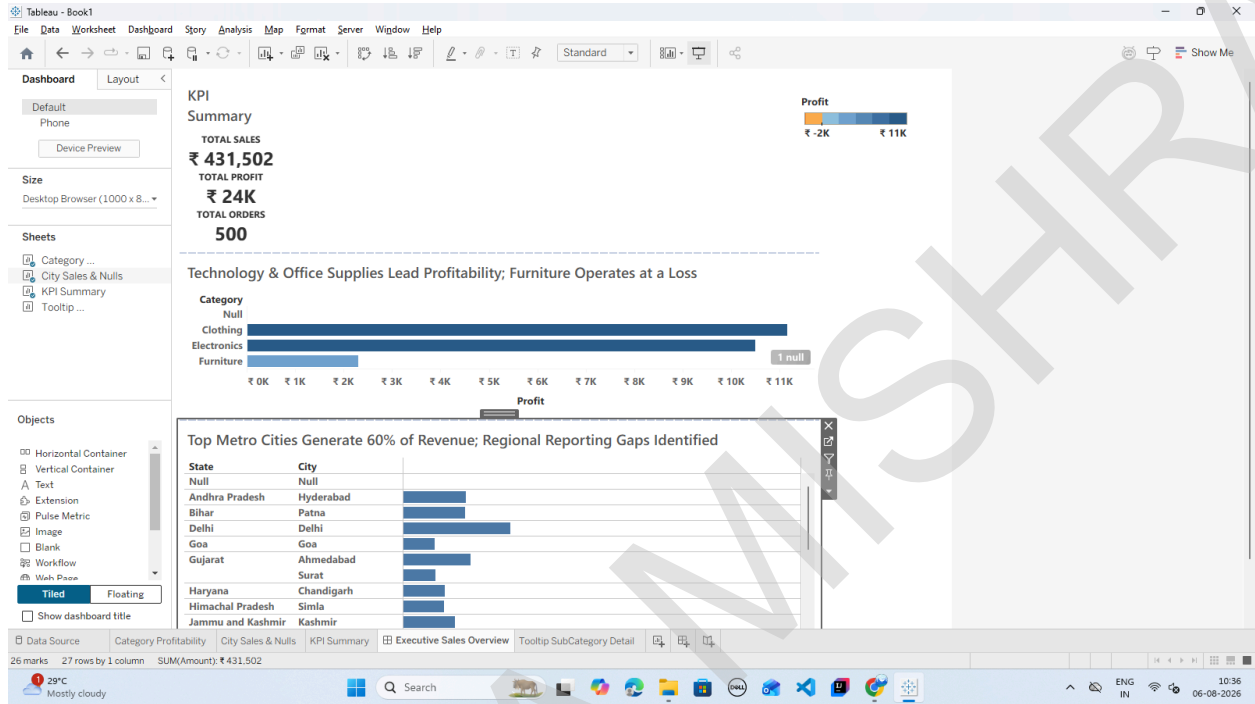


NAME: AASHKA MISHRA  
ROLL NO: T095

SHETH L.U.J. AND SIR M.V. COLLEGE  
SUBJECT: Data Visualization with Power BI and Tableau

Task 2: Replace Descriptive Headers with Action-Oriented Insight Titles

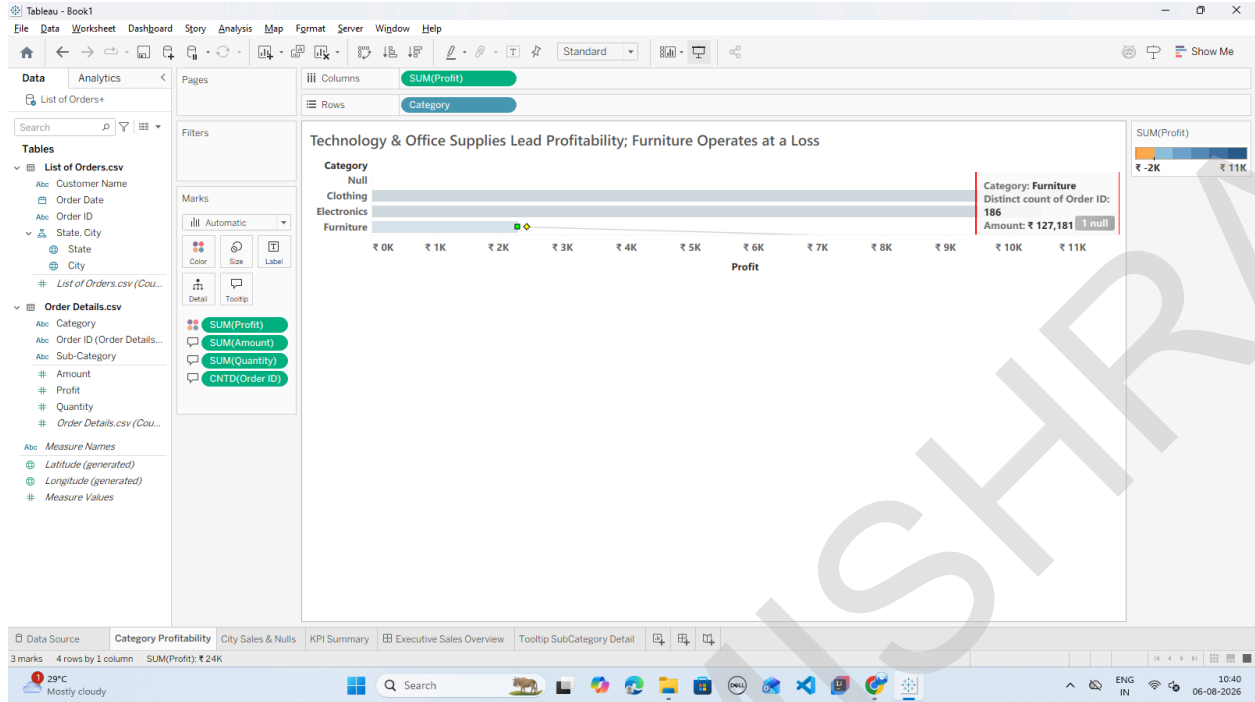
1. Double-click the title block of Category Profitability on your dashboard.
2. Edit the title text to be insight-driven:
3. Double-click the title block of City Sales & Nulls.
4. Edit the title text:



Task 3: Add Direct Data Annotations for Significant Points

1. Right-click on the bar with the lowest profit (or highest loss) in your Category Profitability sheet.
2. Hover over Annotate  $\rightarrow$  select Mark.
3. In the annotation window, write a brief explanatory callout:
4. Format the text to 9pt, Dark Gray, Bold, then click OK.
5. Drag the line callout box so it sits cleanly above or next to the bar without overlapping neighboring data.

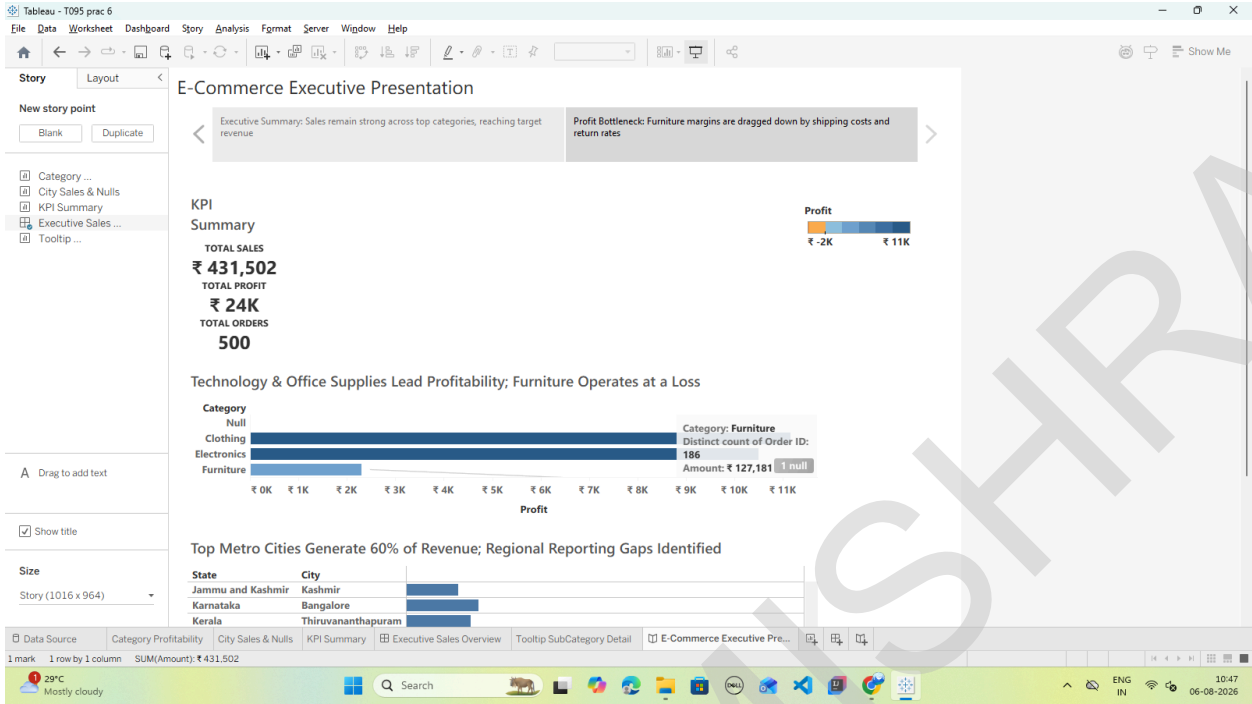
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SUBJECT: Data Visualization with Power BI and Tableau



**Task 4: Build a Guided Story Sequence (Tableau Story Points)**

1. At the bottom of your Tableau window, click the New Story icon (the book icon next to New Dashboard).
2. Name the story tab: E-Commerce Executive Presentation.
3. In the left-hand Story Pane, drag your Executive Sales Overview dashboard onto the canvas.
4. Click on the gray navigation box above the view (the "Story Point" caption box) and type your first story point heading:
5. In the top-left pane, click Blank under New Story Point to add a second slide.
6. Drag Executive Sales Overview onto the canvas again.
7. On the dashboard view itself, click on the Furniture category bar (or lowest profit bar) to filter/highlight it across the entire screen.
8. Click the caption box for Slide 2 and type:
9. Click Save on your workbook (Ctrl + S / Cmd + S).

SHETH L.U.J. AND SIR M.V. COLLEGE  
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